



Aror University of Art, Architecture, Design and Heritage

Faculty of Artificial Intelligence and Multimedia and Gaming

Work Sheet: 03

Q: 01 Use a calculator to evaluate each function at the given value of x .

$$^x, x = 3$$

1. $f(x) = 0.3$

$$^x, x = 1.5 \quad 2. f(x) = 30$$

$$^x, x = 2 \quad 4. f(x) = \left(\frac{1}{2}\right)^{2^x}, x = \pi$$

3. $f(x) = -2$

$$^x, x = -2.5 \quad 5. h(x) = 2 - x^{-x^2}, x = 0.5$$

4. $x(x) = x$

Q: 02 Sketch the graphs of the following exponential functions.

1

$$_2)^{-x} + 3 \quad 3. x(x) = 5^{x-2} + 4$$

1. $f(x) = 4^{-x} + 4 \quad 2. x(x) = ($

4. $h(x) = 2 - x^{-x^2} \quad 5. x(x) = 4^x$

x

$$^{3^x-5} \quad 6. x(x) = \log$$

$$_3 7. x(x) = \ln x + 6$$

Q: 03 Use a One-to-One Property to solve the equation for x .

1. $\left(\frac{1}{3}\right)^{x-3} = 9 \quad 2. 3^{x+3} = \frac{1}{81} \quad 3. x^{3x-5} = x^7 \quad 4. x^{8-2x} = x^{-3}$

5. $\log_4(x+7) = \log_4 14 \quad 6. \log_8(3x-10) = \log_8 5 \quad 7. \log(x^2+6x) = \log 278. \ln(x-9) = \ln 4$

Q: 04 Describe the transformation of the graph of f that yields the graph of g . $x + 2$

$$^x, g(x) = 5$$

1. $f(x) = 5$

$$^{x+1}$$

$$^x, g(x) = 6$$

2. $f(x) = 6$

x

3. $f(x) = 3^x, g(x) = 1 - 3$

$$4. \log_2(x) = \frac{1}{x+2}$$

$$5. \log(x) = \log(x), \log(x) = 4 - \log(x - 2)$$

$$6. \log(x) = \ln(x), \log(x) = \ln(-x) - 2$$

Q: 05 Write the exponential equation in logarithmic form.

$$1. 3^3 = 27 \quad 2. 25^{32} = 125 \quad 3. x^0 = 1$$

Q: 06 Evaluate each logarithm at the given value of x.

$$1. \log(x) = \log(1000) \quad 2. f(x) = \log_9(x) = -3 \quad 3. f(x) = \log_3(x) = 181$$

$$4. \log_2(x) = \frac{1}{2} \ln(x), x = x$$

Q: 07 Use the properties of logarithms to simplify the expressions.

$$1. \log_8 8 \quad 2. \log_8(x^2) \quad 3. \log_{7.5} 1 \quad 4. 5^{\log_5 3} \quad 5. \ln(x) \quad 6. \ln(x) \cdot 7. \ln(x)$$

logarithm of a single quantity.

$$1. \frac{1}{3} [\log_8(x) + 2 \log_8(x + 4)] - \log_8(x - 1) \quad 2. \ln(x) - [\ln(x + 1) + \ln(x - 1)]$$

Q:08 Condense the expression to the algebraically. Approximate the result to three decimal places, if necessary.

$$1. x^{2-3} = x^{-1}$$

$$2. 4(3^x) = 20 \quad 3. 5^x + 8 = 26 \quad 4. 4^{-3x} = 0.10 \quad 7. 3^{2-x} = 400 \quad 6. 3 - 4 \ln(x) = 11 \quad 7. \ln(x) + \ln(x + 1) = 29$$

$$8. \log(x) = 29 \quad \ln(x + 1) = 2 \quad \ln(x) = 2 \quad 10. 8x^{-2 \cdot 3} = 11 \quad 11. -x^{-x} + x^{-x} = 0 \quad 12. 4^{2x-7} = 64 \quad 13. x^{2x} - 5x^x + 6 = 0$$

Q:10 Students in a mathematics class took an exam and then took a retest monthly with an equivalent exam. The average scores for the class are given by the human memory model

$$f(t) = 80 - 17 \log(t + 1), 0 \leq t \leq 12$$

Where t is the time in months.

(a) Use a graphing utility to graph the model over the specified domain.

- (b) What was the average score on the original exam ($t = 0$)? (c) What was the average score after 4 months?
- (d) What was the average score after 10 months?

Q:11 The number V of computers infected by a virus increases according to the model $V(t) = 100e^{4.6052t}$, where t is the time in hours. Find the number of computers infected after (a) 1 hour (b)

1.5 hours (c) 2 hours.